RATUN RAHMAN

MACHINE LEARNING & NLP RESEARCHER | LLMs, RAG, GRAPH REASONING & TRUSTWORTHY AI

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linkedin.com/in/ratun-rahman-46762b187 | Ph.D. expected December 2026 | Open to U.S. remote/hybrid and relocation

RESEARCH PROFILE

Ph.D. candidate in Computer Engineering and AI/ML researcher developing reliable learning methods for ambiguity, conflicting evidence, heterogeneous data, and resource constraints. Research spans large language models, retrieval-augmented generation (RAG), structured generation, graph reasoning, semantic representation learning, multimodal AI, probabilistic modeling, and federated/distributed learning. Author of a NeurIPS 2026 main-track paper and peer-reviewed IEEE research; experienced in building reproducible machine-learning pipelines, benchmarking against controlled baselines, evaluating robustness, and communicating research from design through publication.

TECHNICAL SKILLS & RESEARCH METHODS

Machine Learning & LLMs: PyTorch, TensorFlow, scikit-learn, Transformers, LoRA / parameter-efficient fine-tuning, semantic embeddings, contrastive learning, ranking objectives, structured prediction, RAG and information retrieval.

Methods: Graph-based reasoning and message passing; probabilistic/Bayesian methods; multimodal learning and fusion; federated learning, personalization, non-IID learning, uncertainty and robustness analysis.

Evaluation: Benchmark development, ablation studies, controlled comparisons, failure-mode analysis, statistical evaluation, sensitivity studies, experiment tracking, data preprocessing and reproducibility.

Programming & Compute: Python, C++, Rust, NumPy, SciPy, NetworkX, CUDA, Linux, Git/GitHub, GPU/HPC workflows; research prototyping and technical writing.

EDUCATION

Ph.D. in Computer Engineering | University of Alabama in Huntsville | Aug 2023–Dec 2026 (expected)

GPA: 3.83/4.00. Research in ML/AI, distributed and quantum learning, resource-efficient computation, and model reliability.

B.Sc. in Software Engineering | Islamic University of Technology | Jan 2018–May 2022

GPA: 3.62/4.00. Focus: security and machine learning.

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant | Network Intelligence & Security Lab, University of Alabama in Huntsville | Aug 2023–Present

- Conduct research in structured generation, LLM evaluation, retrieval grounding, federated/personalized learning, probabilistic ML, multimodal modeling, and reliable computation.
- Build and evaluate Python/PyTorch ML pipelines for data processing, graph-based candidate modeling, embedding analysis, learning algorithms, and benchmark-based comparisons; organize experiments for reproducibility.
- Design robust experimental protocols with baseline comparisons, ablation and sensitivity studies, statistical analysis, and investigation of model failure modes under ambiguity, noise, or heterogeneous data.
- Write and present peer-reviewed research; collaborate across academic research teams and mentor junior researchers on implementation, benchmarking, and scientific writing.

Quantum Security Research Intern | King Abdullah University of Science and Technology (KAUST), Saudi Arabia | Jan 2026–Jun 2026

- Developed a hybrid kernel-based malware-family classifier using supervised feature projection, Nyström approximation, and multiclass ridge regression, evaluated on 18,836 samples spanning 23 classes.
- Compared against classical learning baselines and performed finite-sample/finite-shot evaluation, statistical comparisons, and simulator-versus-hardware validation; manuscript under revision at IEEE TDSC.

Game Developer | Alienide Interactive, Dhaka, Bangladesh | Jun 2022–May 2023

- Developed and tested interactive applications using Unity, C#, Python, Blender, and AR/VR tools; worked across implementation, debugging, and iterative feature delivery.

SELECTED LLM, NLP & AI RESEARCH

Ambiguity-Aware Structured Generation | NeurIPS 2026, Main Track — Accepted

- Formulated structured generation as ambiguity-aware inference over multiple semantically valid outputs, rather than single-reference prediction.
- Constructed candidate representations as graphs of entities and relationships; applied semantic-alignment and relational-consistency reasoning to select among plausible interpretations.
- Evaluated accuracy, stability, and robustness on structured-generation benchmarks including nvBench and Spider under differing levels of ambiguity. Paper: arxiv.org/abs/2610.31681.

Interference-Aware Retrieval-Augmented Generation | RAG / Faithfulness | Manuscript under review, 2026

- Investigated evidence aggregation when retrieved documents conflict, overlap, or contain misleading support; modeled interactions among passages and answer candidates instead of treating evidence as independent.
- Designed evaluation around faithful generation, conflicting evidence, and failure analysis for retrieval-based question answering and answer synthesis.

AlignSim: Semantic Representation and Contrastive Learning | Research prototype, 2026

- Built LoRA-based representation-learning experiments using CoSENT, InfoNCE, ranking, band, and geometry-inspired objectives to study semantic alignment.
- Created challenge sets testing synonymy, antonymy, negation, word-order changes, and robustness of sentence representations.

Semantic Communication-Aware Federated Fine-Tuning of LLMs | IEEE Communications Letters, 2025

- Coauthored work on communication-aware federated LLM fine-tuning, using compact semantic representations and coordination under distributed-client constraints.
- Studied fine-tuning and communication-efficiency trade-offs with text-learning tasks including WikiText-2 and AG News.

FEDERATED, PROBABILISTIC & MULTIMODAL MACHINE LEARNING

Probabilistic Federated Learning with Personalization | IEEE TETCI, 2026

- Developed methods for uncertainty-aware, personalized federated learning in heterogeneous and non-IID environments, combining probabilistic reasoning and client-specific adaptation.
- Evaluated predictive learning and model personalization with image-classification settings including CIFAR-10, CIFAR-100, and Tiny ImageNet.

Multimodal Federated Learning for Air-Quality Estimation | IEEE GRSL, 2025; NeurIPS OPT Workshop, 2024

- Co-developed approaches to fuse heterogeneous aerial and ground measurements while personalizing learning across decentralized participants.
- Analyzed modality heterogeneity, multimodal fusion, local adaptation, and collaborative prediction under distributed-data constraints.

Personalized Forecasting & Distributed Prediction | IEEE IoT Journal, 2025; IEEE TVT, 2024

- Developed federated approaches for electrical-load prediction across multihop smart-meter networks and personalized recognition under non-IID data.
- Studied local-versus-global learning trade-offs, communication constraints, and predictive performance in distributed infrastructure settings.

Graph-Based Visual Reasoning | Ongoing research, 2026

- Investigating question-conditioned graph decomposition and probabilistic approximation for compositional visual reasoning as an ongoing research direction.
- Using visual-question-answering benchmarks to study reasoning accuracy, decomposition quality, and compute/accuracy trade-offs.

SELECTED PEER-REVIEWED AI/ML PUBLICATIONS

- Ratun Rahman and Atit Pokharel. “Rethinking Structured Generation: Can Graph-Based Reasoning Resolve Ambiguity?” **NeurIPS (main track)**, 2026.
- Ratun Rahman and Dinh C. Nguyen. “Probabilistic Federated Learning on Uncertain and Heterogeneous Data with Model Personalization.” **IEEE Transactions on Emerging Topics in Computational Intelligence**, 2026.
- Samia Rahman and Ratun Rahman. “Semantic Communication-Aware Federated Fine-Tuning of Large Language Models.” **IEEE Communications Letters**, 2025.
- Ratun Rahman, Falguni Patadia, Jinhui Wang, and Dinh C. Nguyen. “Multimodal Federated Learning for Air Quality Estimation with Model Personalization over Aerial-Ground Networks.” **IEEE Geoscience and Remote Sensing Letters**, 2025.
- Ratun Rahman, Pablo Moriano, Samee U. Khan, and Dinh C. Nguyen. “Electrical Load Forecasting over Multihop Smart Metering Networks with Federated Learning.” **IEEE Internet of Things Journal**, 2025.
- Ratun Rahman and Dinh C. Nguyen. “Improved Modulation Recognition Using Personalized Federated Learning.” **IEEE Transactions on Vehicular Technology**, 2024.
- Ratun Rahman and Dinh C. Nguyen. “Multimodal Federated Learning with Model Personalization.” **NeurIPS OPT Workshop**, 2024.
- Ratun Rahman and Dinh C. Nguyen. “Electrical Load Forecasting in Smart Grid: A Personalized Federated Learning Approach.” **IEEE CCNC**, 2024.
- Md. Raihan Uddin, Ratun Rahman, and Dinh C. Nguyen. “False Data Injection Attack Detection in Edge-Based Smart Metering Networks with Federated Learning.” **IEEE CCNC**, 2024.

- Ratun Rahman, M. R. Islam, A. Ahmed, M. K. Hasan, and H. Mahmud. “A Study of Permission-Based Malware Detection Using Machine Learning.” **International Conference on Security of Information and Networks**, 2022.

ADDITIONAL RELEVANT PUBLISHED RESEARCH

- Ratun Rahman and Dinh C. Nguyen. “Quantum Noise Mitigation with Adaptive Zero-Noise Extrapolation: A Contextual Multi-Armed Bandits Approach.” **IEEE Journal on Selected Areas in Communications**, 2026. Contextual-bandit decision making; evaluation on superconducting quantum hardware.
- Ratun Rahman, Shaba Shaon, and Dinh C. Nguyen. “Tackling Heterogeneity in Quantum Federated Learning: An Integrated Sporadic-Personalized Approach.” **IEEE Transactions on Computers**, 2026.
- Ratun Rahman and Dinh C. Nguyen. “Quantum Convolutional Neural Networks: A Survey on Architectures, Applications, and Future Directions.” **IEEE Transactions on Neural Networks and Learning Systems**, 2026.

SELECTED MANUSCRIPTS & RESEARCH IN PROGRESS

RAG / LLM evaluation: “When Evidence Conflicts: Interference-Aware Aggregation for Faithful Retrieval-Augmented Generation.” Under review, 2026.

ML security: “Scalable Malware Family Classification Using Quantum Kernel-Based Machine Learning.” IEEE Transactions on Dependable and Secure Computing, under revision, 2026.

Efficient learning: “On Resource-Efficient Quantum Convolutional Neural Networks with Provable Convergence Guarantees.” IEEE TNNLS, revision 1 submitted, under review, 2026.

RESEARCH TOOLS, BENCHMARKS & EXPERIMENTAL PRACTICE

Structured language / reasoning: Worked with structured-generation benchmarks nvBench and Spider; evaluated candidate-level ambiguity, semantic alignment, and relational consistency using controlled comparisons and robustness analysis.

LLM and semantic learning: Fine-tuning and representation-learning tasks involving WikiText-2, AG News, semantic similarity, negation, paraphrase and lexical/ordering perturbations.

Distributed / probabilistic learning: Image and multimodal settings including CIFAR-10, CIFAR-100, Tiny ImageNet, aerial-ground sensor observations, smart-grid measurements and heterogeneity-aware personalization.

Practical ML evaluation: Compared statistical and machine-learning baselines, conducted ablations and repeat-run comparisons, documented assumptions, and analyzed the cost of model complexity and data or compute constraints.

ADDITIONAL RESEARCH & ENGINEERING CONTRIBUTIONS

Adaptive Decision-Making Under Resource Constraints | IEEE JSAC, 2026

- Developed contextual-bandit optimization for choosing among adaptive error-mitigation strategies based on measurement uncertainty, estimated benefit, and execution cost; evaluated 240 Rigetti superconducting-QPU tasks through Amazon Braket.
- Transferable strengths include sequential decision making, cost-sensitive policy selection, experiment design, and robustness evaluation.

Reproducible Research Software & Technical Communication | 2023–2026

- Created reusable simulation and evaluation workflows with configurable parameters, measurable benchmarks, numerical visualizations, and clear research documentation.
- Communicated technical methods through conference papers, technical reports, research presentations, and peer review; collaborated with multidisciplinary teams.

RECOGNITION, SERVICE & LEADERSHIP

- NeurIPS 2026 main-track accepted poster, structured generation and graph-based reasoning.
- Alabama EPSCoR Graduate Research Scholars Program (GRSP) Fellowship, 2026; Mary Makima and Lester Ross Scholarship, UAH, 2026.
- International Collegiate Programming Contest (ICPC) Bangladesh Finalist, 2020; IUT Programming Competition Bronze Prize, 2019.
- Peer review service including NeurIPS 2025, ICLR 2026, IEEE journals, ACM Computing Surveys and ACM TKDD; mentoring and reproducibility support for research teams.
- Technical service: lead web developer for an NSF-funded Research Experiences for Undergraduates (REU) website at UAH.